

Artificial Intelligence and Knowledge Engineering Assignment 2

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1 Goal

This assignment is a practical introduction to game playing algorithms for two player full information games - Minimax algorithm and its improvement through alpha-beta pruning.

1.1 Definitions

Minimax Algorithm is a strategy for searching a game tree, used for decision-making in two-player zero-sum games such as chess, draughts, or Go. The algorithm aims to minimise the maximum possible loss a player may incur, taking into account the opponent's moves.

The Minimax algorithm is based on traversing a game tree, where each node represents a game state and edges correspond to players' moves. The leaves are nodes representing terminal states of the game — that is, a win for one of the players. In practice, due to the number of nodes growing exponentially with depth, a **depth limit** and **heuristic evaluation functions** are used to assess non-terminal game states when this limit is reached.

Alpha-beta Pruning The Minimax algorithm can be improved using alpha-beta pruning, which consists in terminating the recursive exploration of certain branches of the tree when it can be determined that further exploration will not affect the final decision.

2 Tasks

Breakthrough is a board game designed by Dan Troyka in 2000. A two-player game is played on a square, chessboard-like grid with a default size of 8×8 . Each player has their own colour, typically white and black. At the start of the

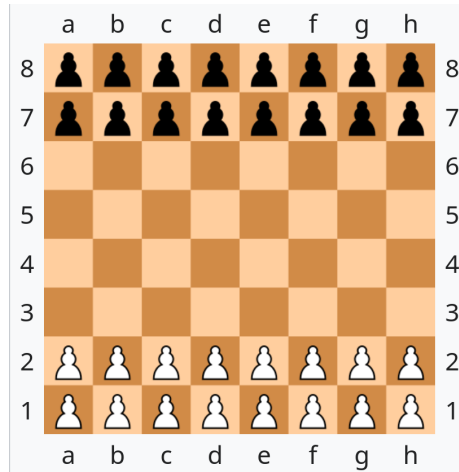


Figure 1: Breakthrough – initial board configuration.

Source: Source: [https://en.wikipedia.org/wiki/Breakthrough_\(board_game\)](https://en.wikipedia.org/wiki/Breakthrough_(board_game))

game, each player occupies the two rows closest to them with their pieces, as shown in Fig. 1. The white player moves first.

Players take turns alternately. A player may move one of their pieces one square forward or diagonally, provided that the target square is empty. A player may capture an adjacent opponent's piece and occupy its square only if the opponent's piece is located diagonally relative to the moving piece. Capturing is not possible when pieces are directly in front of each other. The winner is the player who first moves one of their pieces to the opposite edge of the board. For instance, in Fig. 1, the white player must reach any square in the eighth row to win the game.

Using the material provided in this assignment and during the lecture, write a program that plays the game Breakthrough on a parametrised board size $n \times m = 8 \times 8$ (it may be modified). The program should accept the initial board configuration from standard input (e.g. the default starting configuration of Breakthrough), allow the selection of a heuristic, and enable adjustment of the parameter d , which denotes the maximum search depth in the decision tree. The board representation should consist of m lines, each containing n space-separated symbols: B denotes a piece of the first player, W denotes a piece of the second player, _ denotes an empty square, and o denotes the square from which the last move was made.

It is now the first player's turn. After execution, the program should:

- Basic version: play a complete game using the Minimax algorithm from the perspective of the first player. The second player always makes optimal moves (both players use the same heuristics, hence the first player's predictions of the opponent's moves are always accurate).

- Extended version (additional 20 points): allow gameplay between two independent agents (the opponent does not have to act optimally, and players may change strategies).

The program should output to standard output the board representation after the game has finished. In an additional line, it should report the number of rounds played and indicate which player has won. To standard error, the program should output the number of visited nodes in the decision tree and the execution time of the algorithm.

Marking Scheme:

1. Correct definition of the game state and a function generating possible moves for a given state and player. (10 points)
2. Construction of a **set of heuristic evaluation functions** for each player; each player should have at least three different strategies. (20 points)
3. Implementation of the **Minimax** method and the program in its basic version. (30 points)
4. Implementation of **alpha-beta pruning** in the basic version. (40 points)
5. Extension of the program to the advanced version, enabling gameplay between two programs (e.g. two instances of the same application) and the use of heuristics that adaptively change the player's strategy in order to achieve victory (**additional 20 points**).

Prepare a report for the assignment containing a theoretical description of the method, a formal formulation of the problem including constraints, a definition of the game state and the decision tree, and a description of the solution approach together with a brief example. The report should also indicate the sources used and briefly describe any libraries employed in the implementation. In the conclusion, additionally describe any implementation issues encountered during the task. The report should be submitted to the instructor at least 24 hours before the assignment deadline.

3 Literature

- Breakthrough - Wikipedia
- Breakthrough - opis zasad
- J.F. Nordstrom – Introduction to Game Theory
- M. Maschler et al., Game Theory
- N. Nisan et al., Algorithmic Game Theory
- M. J. Osborne, An Introduction to Game Theory